

JACOB COOK

Information Security Engineer | Detection · IR · Pen Testing · Vulnerability Mgmt | Linux · Firewall · Cloud · EDR/NDR

Sacramento, CA · 916 709 1794 · jacobcookofficial@gmail.com
[linkedin.com/in/jcook-dev](https://www.linkedin.com/in/jcook-dev) · github.com/jacobdcook · jacobdcook.com

Hands-on security engineer with breadth across threat detection, incident response, pen testing, and vulnerability management. Built a Wazuh SIEM from scratch on Linux infrastructure, authored Sigma rules mapped to MITRE ATT&CK, and engineered Python/FastAPI automation for response workflows. CompTIA-certified across four disciplines (Security+, CySA+, PenTest+, SecurityX); BS Computer Science; MS Cybersecurity in progress. Strong analytical and problem-solving skills, with detail-and-follow-through discipline applicable to regulated environments.

TECHNICAL SKILLS

Detection & IR: Wazuh SIEM (build & admin), Sigma Rule Authoring, MITRE ATT&CK, EDR/NDR Concepts, IOC Analysis, IR Lifecycle, Microsoft Sentinel (familiar), KQL/SPL (familiar), Threat Hunting

Systems & Networking: Linux (Ubuntu / CentOS) Build & Admin, Unix / macOS, Windows Endpoints, Firewall Concepts & Policy Awareness, Wireshark, Networking (TCP/IP, DNS, SSL/TLS)

Pen Testing & Vulnerability Mgmt: CompTIA PenTest+ certified, Vulnerability Assessment Concepts, OWASP, System Hardening, Remediation Coordination

Cloud, Identity & Dev: Azure AD / Entra ID, Okta, AWS labs, SSO / RBAC, Terraform, DevSecOps Practices, Python · Bash · SQL · FastAPI · Git

TECHNICAL PROJECTS

Blue Team SOC Monitoring Lab: Wazuh SIEM (End-to-End Linux Build) 2026

- Built a full Wazuh SIEM stack from scratch on Linux — deployment, log ingestion, normalization, alert rules — direct hands-on with Linux operations and security technology systems.
- Validated 12 detection patterns against 242 simulated events with 100% precision after tuning, reducing false-positive volume by 100% by suppressing 79 benign-only matches.
- Verification: 19 passing pytest cases in repo (tests/test_detection_accuracy.py).

Stryker / Intune Detection Pack: Detection-as-Code 2026

- Authored 6 Sigma-format detection rules mapped to MITRE ATT&CK that eliminated 100% of false positives in a 100-event simulation using documented baseline-lookup logic.
- Documented 18 benign scenarios as repeatable evidence — the same audit-ready discipline regulated environments require.
- Verification: 43 passing pytest cases in repo (tests/test_rules.py).

SOAR-lite: Incident Response Orchestrator (Python / FastAPI) 2026

- Built end-to-end IR automation in Python/FastAPI — webhook ingestion, playbook routing, IP blocking, Slack notification, and incident write-up — reducing triage time by over 60% in lab testing.
- Eliminated four manual triage steps with documented runbook coverage.
- Verification: 10 passing pytest cases in repo (tests/test_orchestrator.py).

Phishing Analysis Lab 2026

- Triageed 5 phishing samples end-to-end with header inspection, SPF/DKIM/DMARC validation, IOC extraction, and attacker infrastructure mapping — 100% accurate verdicts.

G3-GPT: Industry-Sponsored Enterprise AI Platform (Team of 6) 2024

- Engineered RBAC and Azure AD SSO controls on a production AI platform — direct cloud security and IAM implementation work in a regulated-style sponsor environment.
- Shipped enterprise-grade platform on deadline under direct industry sponsor oversight with zero critical security findings at delivery.

PROFESSIONAL EXPERIENCE

Target Distribution Center | Operations Engineer: Systems Monitoring & Incident Response 2019 - 2025

- Maintained 99.5% uptime in a high-paced distributed-systems environment — direct experience operating in a fast-moving operational context with strong attention to detail.
- Resolved 200+ anomalies through structured root cause analysis with tracked documentation that reduced repeat incidents by an estimated 30% — applied analytical and problem-solving discipline transferable to financial-services-grade environments.
- Reduced mean time to recovery by an estimated 20% via cross-team coordination, escalation, and post-incident reporting.

EDUCATION & CERTIFICATIONS

WGU | MS, Cybersecurity (In Progress)

Expected Oct 2026 · GPA 3.50

CSUS | BS, Computer Science

Jan 2025 · GPA 3.50

CompTIA: Security+ · CySA+ · PenTest+ · SecurityX · CSIE (stackable)